HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · CONSUMER DURABLES MANUFACTURING · PROCUREMENT

ROOM AIR CONDITIONERS · STRATEGIC SOURCING · SUPPLIER EVALUATION MATRIX

Choose suppliers on evidence

A room-AC manufacturer with hundreds of suppliers behind every unit now ranks them on one weighted matrix its category heads signed, backed by segmentation, a volume split optimiser, a disruption simulator and GenAI that reads quotes.

~~3 wks~~ → **2** days per supplier evaluation cycle



2 days
Supplier evaluation cycle
▼ from about 3 weeks



38 % lower
Incoming defect rate, key suppliers
scorecards drive improvement plans



3.2 %
Material cost saving, evaluated categories
landed cost + volume splits



1 left, from 3
Single-source critical parts
second sources developed

Rings: 2 days against a 21-day cycle; the 38% reduction; the eight criteria behind the saving; the 3 single-source parts, 2 now closed (teal), 1 left (orange). Figures indicative; validated metrics available on request.

About the charts: sections 01, 02, 04 and 05 are drawn from the case-study workspace's synthetic supplier base for a 1.5-ton inverter split AC: 8 component families, 12 anonymised suppliers across India, China, Thailand and Vietnam. Illustrative only.

01 Agree the weights, then let the scores rank

12 suppliers × 8 criteria · balanced weight preset

RANK · SUPPLIER	Quality wt 20%	Delivery wt 18%	Cost wt 18%	Capacity wt 8%	Financial wt 10%	Lead time wt 8%	ESG wt 8%	Local wt 10%	Weighted score balanced preset
1 Supplier D · India — Copper tube & coil	81	80	50	66	84	85	80	95	76
2 Supplier K · India — Sheet metal & mouldings	76	85	55	26	88	90	83	100	75
3 Supplier I · India — BLDC motors	65	70	46	33	79	78	72	80	64
4 Supplier H · India — Inverter PCBs	49	65	27	99	66	73	74	70	60
5 Supplier B · Thailand — Compressors	74	75	36	55	81	43	78	0	57
6 Supplier F · India — Aluminium fin stock	60	45	41	18	61	82	59	100	56
7 Supplier E · Vietnam — Copper tube & coil	70	50	64	44	70	48	66	0	54
8 Supplier A · China — Compressors	53	55	68	40	72	22	64	0	50
9 Supplier C · India — Compressors	19	40	18	88	58	79	70	85	48
10 Supplier J · China — BLDC motors	43	30	73	66	68	28	52	0	46
11 Supplier G · China — Inverter PCBs	35	35	81	48	75	10	55	0	44
12 Supplier L · India — Mouldings & packaging	24	20	32	44	49	87	47	100	43

Each criterion scored 0–100 from raw metrics (defects per million, on-time-in-full, price index, capacity headroom, lead time…)

0 — 100 criterion score ■ top-ranked supplier

THE METHOD
Eight criteria, one formula
Weights set by pairwise comparison (AHP) with a consistency check, signed off by the category heads before any supplier is scored.

CHEAPEST IS NOT BEST
Supplier G: cost 81, rank 11
The best price score in the table is pulled down by quality 35, delivery 35 and lead time 10.

LOCAL AND RELIABLE
D and K lead the table
Strong quality, delivery and lead time with full local sourcing put both at 75–76.

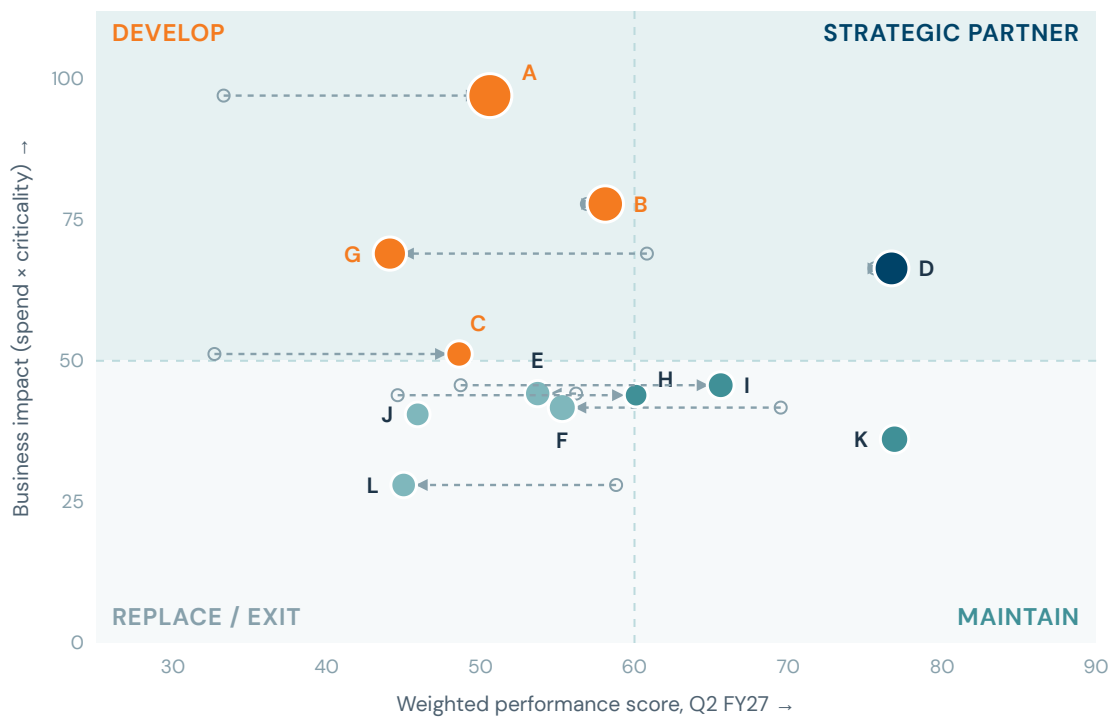
LIVE IN THE TOOL
Presets change the order
Balanced, cost-first, quality-first, resilience and localisation presets re-rank every row, so trade-offs are argued on weights, not on names.

02 Who to grow, who to develop, where the risk sits

Supplier 360 segmentation · component supply risk

Segmentation: performance vs business impact

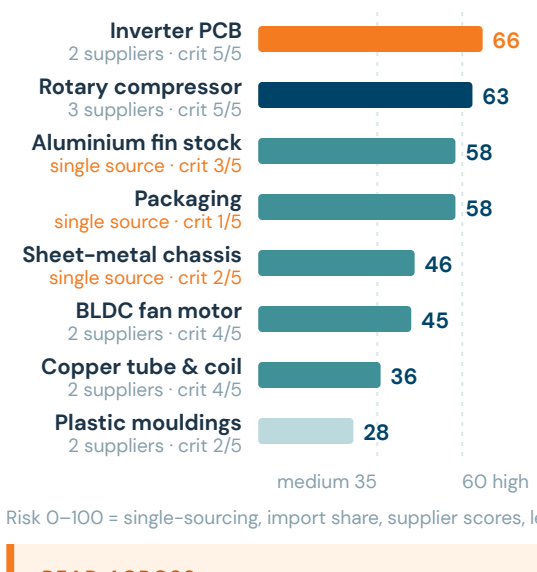
Dot = supplier position in Q2 FY27, size = spend; dashed arrow = move since Q1 FY26 as improvement plans land



Develop: high impact, below target ■ Strategic partner ■ Maintain ■ Replace / exit

Supply risk by component family

Highest risk in orange · from the 3D component explorer



Risk 0–100 = single-sourcing, import share, supplier scores, lead time, criticality

READ ACROSS
Compressors and PCBs carry the risk
The two criticality-5 parts score highest, and their main suppliers (A, G) sit in "Develop": improvement plans with quarterly milestones before any switch.

03 Four pain points, four fixes

Before → after

BEFORE
Data in silos
Quality in the QMS, delivery in the ERP, prices in emails and spreadsheets.

ERP + QMS EXTRACTS

AFTER
One monthly scorecard
Automated per supplier, the same data in every review meeting.

BEFORE
Subjective selection
Rankings depended on who was in the room; weights changed from meeting to meeting.

AHP WEIGHTS

AFTER
Eight criteria, agreed
Partner, develop, maintain, exit segments, reviewed quarterly.

BEFORE
Concentration risk
Critical parts on one supplier or one country; allocation by historical shares; firefighting.

OPTIMISER + SIMULATOR

AFTER
Splits and switches planned
Volume optimised for cost, quality and risk within capacity; cover and cost known.

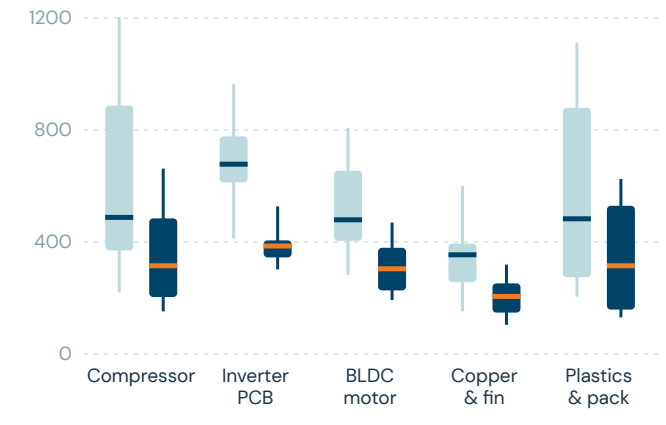
BEFORE
Quotes hard to compare
Currencies, Incoterms and payment terms made landed cost a manual job of days per RFQ.

GENAI QUOTE READER

AFTER
Landed cost in minutes
Quotes in any format normalised; buyers verify every extracted value.

04 Defect spread narrows

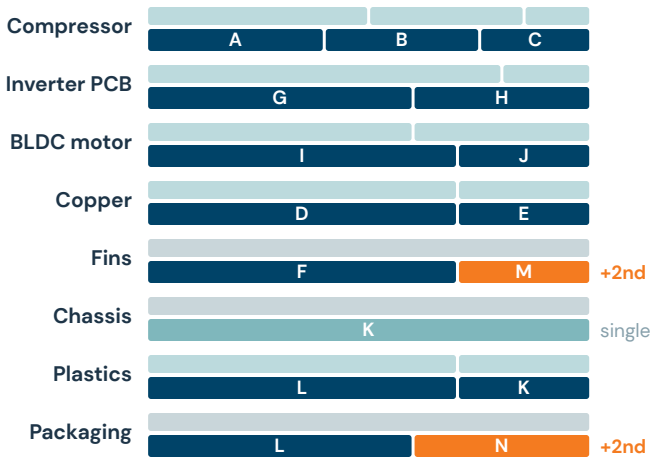
Defects per million by family · before (light) vs after (navy) · box = middle half, line = median



Median across families: 454 → 294 ppm

05 Who supplies what

Supplier share per family · upper bar before, lower bar after · orange = new second source (M, N)



06 How HMC ran it

2 weeks — Discover
Interviews, data sources mapped, single-source exposure listed. Criteria catalogue and data map.

2 weeks — Design
Scoring rules from raw metrics; AHP weights; segmentation rules. Signed-off framework.

5 weeks — Build
Monthly scorecards, optimiser, disruption scenarios, GenAI quote reader, Power BI views.

Pilot categories — Adopt
Quarterly reviews run from the scorecard; RFQs compared on landed cost.

All categories — Scale
Second sources tracked to closure; scorecards shared with suppliers each quarter.

What any procurement team can borrow

01 — Agree the weights before the scores
When the committee signs the weights first, the ranking is accepted, not argued.

02 — Compare on landed cost
The cheapest quote on paper is often not the cheapest at the plant gate.

03 — Design for a second source
Know the cover days and the switch before the disruption, not after.

Built with **ERP & QMS extracts** · **AHP** · **Python** · **Three.js** · **Large language model** · **Power BI** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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